Worksheet: E_k & E_p

Name: _____

Use the kinetic energy and gravitational potential energy formulas in your textbook to answer the following questions - 3 Step Math

1. What is the kinetic energy of a 25kg object moving at a velocity of 5m/s?

2. What is the gravitational potential energy of a 150kg object suspended 5m above the earth's surface?

3. What is the kinetic energy of a 25kg object moving at a velocity of 10m/s?

4. What is the gravitational potential energy of a 2.5kg object that is 300m above the surface of the earth?

5. An object with a kinetic energy of 2160J has a mass of 120kg. What is its velocity?

6. An object whose mass is 43kg is hanging on a thin wire. The object has a potential energy of 3160.5J. How high is the object above the ground?

7. What is the kinetic energy of a 150gram *(* convert to kg)* object moving at a velocity of 100m/s?

8. An object has a kinetic energy of 96J. Its velocity is 4m/s. What is its mass?

9. An object with a kinetic energy of 1125J has a mass of 250kg. What is its velocity?

10. What is the mass of an object that is hanging 12.6m above the surface of the earth and has a potential energy of 2778.3J?

11. An object has a potential energy that is 833 J. Its height above ground is 4.25 m. What is its mass?

12. An object with a kinetic energy of 16,000J has a velocity of 8m/s. What is its mass?

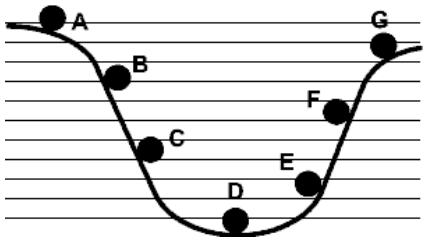
13. An object has a gravitational potential energy of 41772.5J and has a mass of 1550kg. How high is it above ground?

14. What is the kinetic energy of a 25kg object moving at a velocity of 2.5m/s?

Challenge: A+ level.

15. A 2.0kg object is dropped from a height of 30m. After it drops for 2.0 seconds, what is its kinetic energy and what is its potential energy? (Assume no air resistance.)

16. *This graph shows a ball rolling from A to G. The ball starts at point A and rolls to point G.*



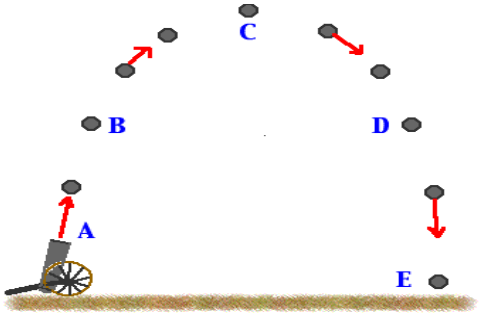
a) At what letter does the ball have the greatest kinetic energy? _____

b) Which letter shows the ball when it has the maximum potential energy? _____

c) Which letter shows the ball when it has the least potential energy? _____

d) Why is point G slightly lower than point A? In other words, why couldn't the ball go back to the same height at which it started?

17. *Use the diagram below to answer the next set of questions*



a) At what letter does the ball have the greatest kinetic energy? _____

b) Which letter shows the ball when it has the maximum potential energy? _____

c) Which letter shows the ball when it has the least potential energy? _____

d) What can be said about the PE and KE at positions B and D? _____
_____
_____
_____

Energy Transformation